Mini Game Hub

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1 External Libraries

- **numpy** - Used for efficient handling of arrays and game board matrix. It simplifies operations on the game board, such as checking valid moves, updating states, and checking the win condition.
- **pygame-ce** - Used for building the graphical user interface of the games. It handles rendering, mouse detection, game loops, and animations.
- **matplotlib** - Used for visualizing game statistics such as top 5 winners and the frequencies of each game
- **time** - Used for implementing delays, timers, and tracking time-based events within games.
- **os** - Used for interacting with the file system via bash (os.system())
- **sys** - Used for taking command line arguments for game.py

2 Design & Architecture

2.1 Authentication System (Bash)

User authentication is handled via shell scripts:

- User credentials are stored in `.users.tsv` with SHA-256 hashed passwords.
- Login allows 3 attempts before temporary lockout.
- Lockout timestamps are stored in `.failed_logins.csv`.

2.2 Leaderboard System

Game statistics are stored in CSV files. The leaderboard script:

- Sorts users by wins, losses, or win/loss ratio
- Formats output using `column`
- Maintains a top-5 players file

2.3 Game Class Abstraction

The `Game` class stores:

- The board as a NumPy 2D array
- Current player turn (boolean)
- Board dimensions and cell size

It provides utility functions such as `switchTurn()`, `valid_move()`, and a pluggable `checkWin()` function passed during initialization, allowing reuse across different games.

2.4 Rendering and Scaling

All rendering is done using `pygame`. Custom scaling functions `scale_h()` and `scale_w()` ensures resolution-independent scaling by mapping coordinates relative to a reference screen size.

Static elements such as boards are pre-rendered onto `Surface` objects and blitted every frame to improve performance.

2.5 Event Handling and Game Loop

Each game runs a main loop that:

- Captures user input (mouse clicks, key presses)
- Updates game state (moves, turn switching, timers)
- Renders updated visuals

Mouse clicks are mapped to board cells, and moves are executed only if valid.

2.6 Win Detection

Each game defines its own `checkWin()`:

- Tic Tac Toe and Connect 4 scan in 4 directions (horizontal, vertical, both diagonals) using NumPy slicing.
- Othello determines the winner by counting pieces when no valid moves remain (using `np.sum()`)

2.7 Animations

Animations are time-based using timestamps:

- Tic Tac Toe: progressive drawing of crosses (line segments) and circles (arc sweep).
- Connect 4: falling discs simulated using $y = \frac{1}{2}gt^2$.
- Othello: flipping discs using `pygame.draw.ellipse()` and expansion.

Jagged drawing effects are implemented via the `JaggedLine` class, which perturbs line segments randomly for a hand-drawn aesthetic. `pygame.draw.arc()` is naturally jagged, which is what inspired the hand-drawn aesthetic.

2.8 Blitz Mode

A `Timer` class manages turn-based countdowns. Each player's timer activates only during their turn. The remaining time is displayed as a dynamically coloured vertical bar, transitioning from green to red as time decreases.

2.9 User Interface Components

Reusable UI elements include:

- `Button`: handles hover, click feedback, and rendering
- `Box`: used for popups and text displays

End-of-game overlays use semi-transparent surfaces to dim the screen and display options like "Play Again" and "Main Menu".

2.10 Othello Move Logic

Valid moves are computed by scanning in all 8 directions from each empty cell. A move is valid if it captures at least one opponent piece bracketed by the current player's pieces. Piece flipping is performed by updating slices of the NumPy array.

2.11 Connect 4 Mechanics

Each column tracks the number of filled cells. When a move is made, the disc is placed in the lowest available row. Falling animation is applied until the disc reaches its final position.

2.12 Flowchart:

Start
↓
Authentication (main.sh)
↓
Select Game (game.py)
↓
Choose Time Control (game.py)
↓
Toss to Decide First Player (game.py)
↓

TicTacToe	Connect4	Othello
play()	play()	play()

↓
Return Result
↓
Leaderboard (leaderboard.sh)
↓
End

3 Features Implemented

3.1 Authentication System

- Supports user registration and login with passwords securely hashed using SHA-256.
- Two-player authentication is required before starting a game.

- The system enforces a 3-attempt login limit per user, after which a temporary time-based lockout is applied.

3.2 Blitz Feature

- A timed gameplay mode where each player must make a move within a specified time limit.
- The remaining time is visually represented using a vertical timer bar which changes colour according to the urgency of move.

3.3 Animations

Visual enhancements implemented to improve user experience:

- Tic Tac Toe: drawing animations for X and O symbols, along with a line animation highlighting the winning combination.
- Othello: coin flipping with a wave-like animation to represent piece conversion visually.
- Connect Four: a falling animation for coins to simulate realistic gravity-based placement.

3.4 Performance Optimization

- Using set in Othello to improve performance
- Storing data in .user_files to avoid processing from history.csv everytime - Time complexity improvement from O(nlogn) to O(u), where n is number of games and u is number of users
- Preprocessing static images/frames onto Surfaces and only blitting at runtime

3.5 Abstractions

- **Developer Point of View**
  - Design elements which are inheritable as a Class Button, Box, Jagged Line etc.
  - Menus - callable as a function, returns the choice
  - Auto-Sizing of Text - Using Binary Search text is automatically fit to Box/Button - we try a font, and get its rect - If it doesn't fit, we try a font size smaller (We do this using binary search between 1 to 100)
- **User Point of View**
  - GameHub folder is clean - system files and user files are hidden
  - Intuitive Menus and usage

4 Directory Structure

```
hub/
├── main.sh
├── game.py
├── leaderboard.sh
├── reset.sh
├── games/
│   ├── tictactoe.py
│   ├── connect4.py
│   ├── othello.py
│   ├── classGame.py
│   └── design_elements.py
├── images/
│   ├── red_disc.png
│   └── yellow_disc.png
├── .user_files/
│   ├── .stats_tictactoe.csv
│   ├── .stats_connect4.csv
│   ├── .stats_othello.csv
│   ├── .users.tsv
│   ├── .user_total_wins.csv
│   ├── .history.csv
│   ├── .top5.txt
│   └── .game_frequencies.csv
├── .system_files/
│   └── .failed_logins.csv
└── report/
    ├── report.tex
    ├── report.pdf
    ├── references.bib
    ├── makefile
    └── images/
```

5 Installation & Setup

```
pip install numpy pygame-ce matplotlib
```

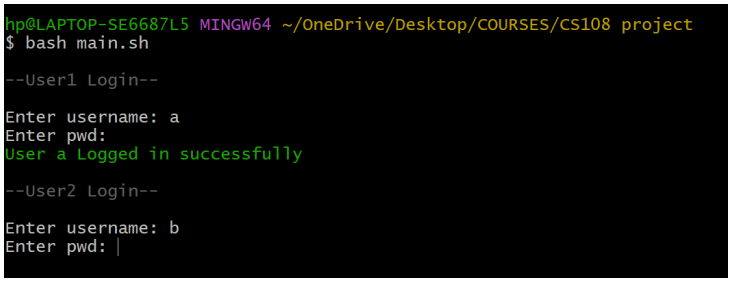
6 Usage

Run the main script to start the application:

```
bash main.sh
```

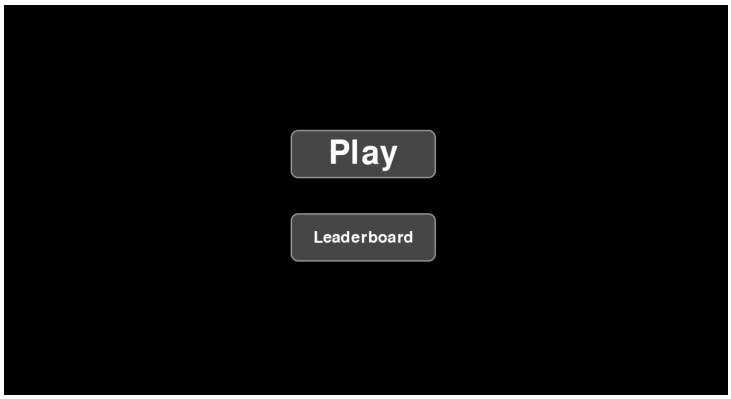
Authentication

Two users are prompted to log in via the terminal. Each user has up to three attempts before a temporary lockout.



Main Menu

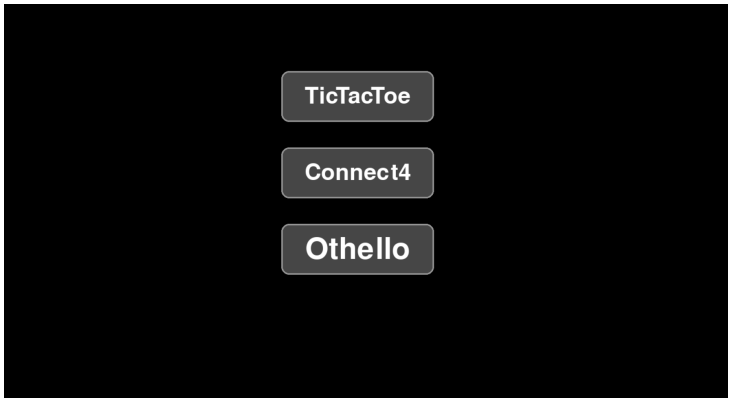
After launching, the main menu is displayed with options to start a game or view the leaderboard.



If play is clicked:

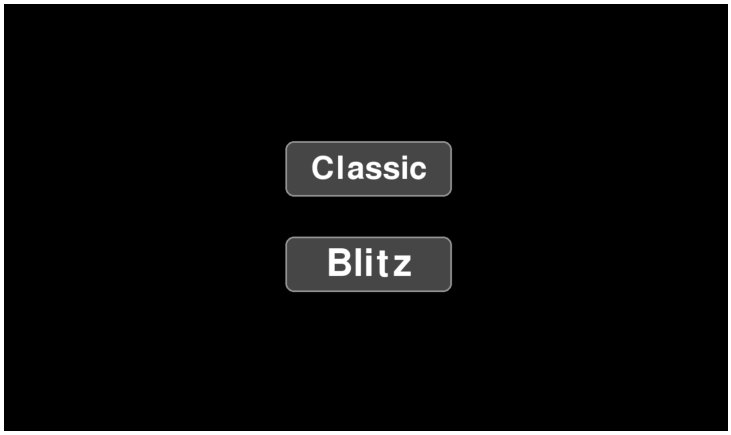
Choosing game

Click the button for game you want



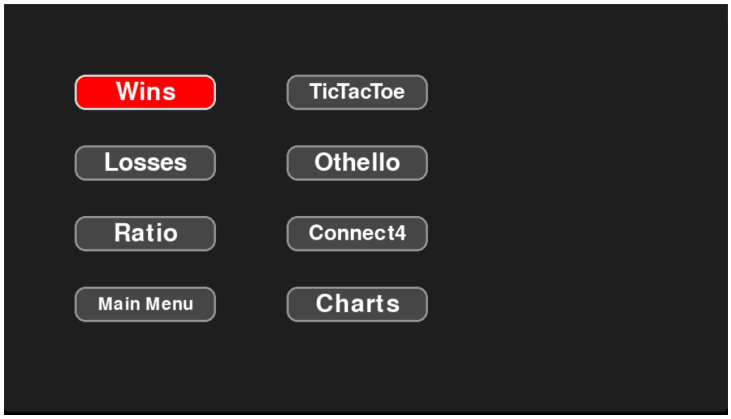
Choosing time control

Click the time control you want



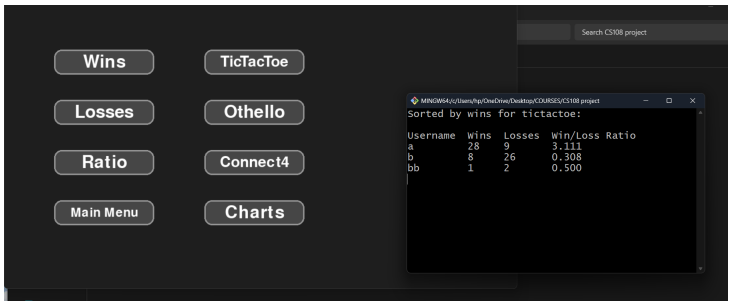
Leaderboard Menu

Click metric to sort by and game name for stats. The first choice is highlighted red until second is chosen



Leaderboard Output

The leaderboard is displayed in the terminal in a formatted table.



7 Retrospective

- Accidental change of the same part by both teammates -
  Fix: Clearly communicating beforehand, and also doing a merge commit when necessary

- Lagging animation -
  Fix: Performance improved by precomputing static frames onto Surface objects and then blitting them to the screen at loop runtime [7]

- Timer and Toss Bar fluid going out of box for low levels -
  Fix: Using pygame Surfaces transparency feature - made a window Surface which lets viewer peek through only the shape we want, hence clipping the fluid [1]

- Scaling of sizes across systems -
  Fix: Using scaling functions wherever necessary

- Not able to change game board values in Othello - the diagonal array extracted from game board using the diagonal() function was not writable by default.
  Fix: Used arr_name.setflags(write=1) [2]

- Seed was same for all board lines since time was converted to int, and execution was finished in less than a second.
  Fix: Used int(time_ns())

- Unnecessary complicated scaling by introducing a Size class -
  Fix: Removed overcomplicated Size class, changed it to simple scaling functions

- Pygame Events during toss() like random clicks and ESCAPE were getting transferred to next loop -
  Fix: `pygame.event.clear()`

8 What we would do differently with 10 more hours

8.1 Code Architecture Improvements

- Introduce a state-machine type flow for game.py to simplify the logic
- Create a Menu Class with Button and Box as subclass
- Use enums for things like whose turn is it to avoid all the confusion during development

8.2 User Experience Enhancements

- Add smoother scene transitions (fade in/out between menus and games).
- Improve aesthetics and feedback (hover effects, button animations, sound effects).
- Add a pause button

8.3 Feature Expansion

- Add single-player mode which just chooses random choices (Just for fun)
- Figure out a cleaner leaderboard display aesthetic so we can see terminal output simultaneously
- Create a visual interface to choose how many seconds per move in blitz

References

[1] *How to Draw a Semi-Transparent Circle in Pygame*. URL: https://stackoverflow.com/questions/51229740/how-to-draw-a-semi-transparent-circle-in-pygame.

[2] *How to Make a NumPy Array Read-Only*. URL: https://www.geeksforgeeks.org/python/how-to-make-a-numpy-array-read-only/.

[3] *Hyperlinks in LaTeX*. URL: https://www.overleaf.com/learn/latex/Hyperlinks.

[4] *Making a Simple Directory Tree*. URL: https://tex.stackexchange.com/questions/5073/making-a-simple-directory-tree.

[5] *NumPy Absolute Beginners Guide*. URL: https://numpy.org/doc/stable/user/absolute_beginners.html.

[6] *Pygame Draw Module Documentation*. URL: https://www.pygame.org/docs/ref/draw.html.

[7] *Pygame Surface Documentation*. URL: https://www.pygame.org/docs/ref/surface.html#pygame.Surface.

[8] *Pygame Transform Smoothscale*. URL: https://www.pygame.org/docs/ref/transform.html#pygame.transform.smoothscale.

[9] *Spines Demo*. URL: https://matplotlib.org/stable/gallery/spines/spines.html.

[10] *Subplots Demo*. URL: https://matplotlib.org/stable/gallery/subplots_axes_and_figures/subplots_demo.html.